Semantic Validation Gates: A Computable, Statistically Calibrated Framework for Runtime Verification of Language-Model Outputs

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Abstract

As language models and autonomous agents produce long reasoning chains and consequential decisions, deployment requires a verification layer that converts qualitative judgments—well-formed, factual, consistent, safe, on-task—into calibrated, auditable measurements. We present a five-gate runtime validation framework in which every object is computable from real model outputs and every guarantee is finite-sample with stated assumptions. The gates are: an exact edit-distance format gate; a calibrated-entailment fact gate with mandatory abstention; a logic gate built on a weighted cellular sheaf Laplacian over the extracted claim graph, unified with propositional constraints through a relax-and-certify architecture whose necessity we prove via a complexity-theoretic incompleteness result; a safety gate that is computationally independent of all parsing; and an intent gate whose aggregation form is forced by a dilution theorem. Thresholds are calibrated by conformal methods under three deployment regimes—exchangeable, covariate-shifted, and arbitrarily (adversarially) drifting—yielding end-to-end, instrument-inclusive false-rejection control. The decision layer evaluates all gates in parallel and grants alignment violations an unconditional veto, provably eliminating the masking of safety failures by lower-priority errors. A pre-registered field study with a temperature-calibrated NLI on real GSM8K reasoning chains locates the sheaf gate's added capability exactly where it is structural: on compositional violations that no sentence pair witnesses, the sheaf energy attains AUROC 0.990 with 1% false-positive rate at 95% recall against chance-level counting baselines, and is the only method with nonzero localization; on pairwise-visible violations it provably coincides with weighted contradiction counting (Prop. 3.5). We position the framework as the missing theory layer for guardrail pipelines already deployed in production without statistical guarantees, and release a reference implementation, the pre-registered protocol, and the field-study harness.

Declaration of generative AI use. The author used a generative AI assistant (Anthropic Claude) in preparing this work: to implement the experimental harness and analysis code for the field study of §6.2, to run and summarize those experiments, and to draft and copy-edit portions of the manuscript text. The research questions, the framework and its architecture, the mathematical claims, the pre-registered protocol and its decision rules, and all interpretations and conclusions are the author's own; the author verified the statements, proofs, and reported numbers and takes full responsibility for the content of this paper. Code, the frozen protocol with its deviation log, and per-run artifacts are released so that every reported number is independently reproducible.

1 Introduction

1.1 Problem

Production systems increasingly interpose *guardrail pipelines* between a language model and its consumers: stacks of validators for output structure, factuality, consistency, safety, and task adherence. These pipelines are effective engineering but carry no mathematical contract: which validator fires first is a configuration accident, error taxonomies are informal, and false-rejection rates are tuned by feel. Conversely, the academic literature supplies deep tools for individual dimensions—conformal factuality, consistency detection, safety classification, instruction-following evaluation—that do not talk to each other. This paper supplies the missing layer: a single formal system in which five validation dimensions are computable scores, thresholds carry finite-sample guarantees under explicitly graded distributional assumptions, the induced error taxonomy is complete and mutually exclusive, and safety violations cannot be masked.

A recurring temptation in this setting is to state the architecture in differential-geometric language—Wasserstein projections onto convex sets of admissible distributions, Čech obstruction classes with Hodge norms, isometric embeddings of Wasserstein geometry into Hilbert spaces. We avoid that route throughout: each such object is either not constructible or not well-posed on the spaces in question, and we say so at the point of use. Every gate below is instead built from an object that exists and can be computed from a real model output.

1.2 Contributions

1. Five gate scores $f_1, \ldots, f_5$ (Format, Fact, Logic, Alignment, Intent), each with an explicit algorithm, estimator, and sensitivity analysis (§3).

2. A logic gate built on a weighted cellular sheaf Laplacian over the extracted claim graph, unified with propositional constraints via relax-and-certify, together with an impossibility result (Prop. 3.4) explaining why a certification layer is unavoidable, and a structural-equivalence result (Prop. 3.5) delimiting exactly when the sheaf energy can add capability beyond weighted contradiction counting (§3.3).

3. Calibration with finite-sample guarantees across three deployment regimes: split conformal under exchangeability, weighted conformal under estimable covariate shift, and adaptive conformal inference whose long-run guarantee holds under arbitrary, including adversarial, drift (§4). The guarantees are instrument-inclusive: claim-extraction and NLI errors cannot inflate the false-rejection rate (Prop. 4.1).

4. A decision architecture with unconditional parallel evaluation and a safety veto, provably immune to masking of alignment violations (§5), together with an anytime computational profile whose safety fast path is a single classifier forward pass (§5.4).

5. A mechanism study on synthetic chains (§6) and its *pre-registered field execution* with a calibrated NLI on real GSM8K chains (§6.2): the field study adjudicates the mechanism's two assumptions, confirms the pre-registered success criteria on compositional violations (AUROC 0.990 vs. chance-level baselines; sole nonzero localization), and shows that on structure-free claim graphs the sheaf energy provably reduces to counting—turning the boundary of the tool's value into a theorem plus a measurement rather than a conjecture.

1.3 Design principles

(P1) Every object is mathematically well-defined and computable from real model outputs. **(P2)** Every guarantee is finite-sample, with its assumptions stated next to it. **(P3)** Signals of localized or safety-critical violations are aggregated by max/veto, never by mean or minimal-index truncation. **(P4)** Anything not meeting P1–P2 is labeled a design choice or an open problem.

1.4 Related work

Each pillar has independent precedents; the contribution here is the certified composition. *Sheaves for consistency:* Huntsman, Robinson and Huntsman [19] demonstrated that LLMs produce usable numerical consistency ratings of claim pairs and proposed sheaf theory for lifting local ratings to global consistency of hypertexts, also noting the relevance of CNF-SAT; our §3.3 differs in operating on a single output's claim graph, weighting the Laplacian by calibrated NLI confidences with a proven sensitivity bound, and embedding the score in a calibrated gate rather than a stand-alone analysis. *Conformal guarantees for LLMs:* conformal factuality [24], conformal abstention for hallucination control [30], boosted and conditional variants [5], and conformal factuality under covariate shift via online density-ratio estimation [7] calibrate a single dimension; §4 calibrates a five-dimensional gate vector jointly with family-wise control and a veto-aware fixed-sequence procedure. *Guardrail pipelines:* production frameworks (Guardrails AI, NeMo Guardrails [25], and similar) and safety classifiers deployed as input–output filters [20] stack validators covering roughly our five dimensions but provide no statistical guarantees, no unified severity measure, and no masking analysis; this paper can be read as the theory layer for that architecture. Multi-dimensional evaluation suites such as HELM [21] systematize offline benchmark dimensions for scoring models; our object is different—runtime, per-output verification with per-instance guarantees and block/repair decisions.

2 Output Representation

Fix a deterministic encoder $E : \Sigma^* \to \mathbb{R}^d$ applied at unit level (in the experiments below, a sentence-embedding model in the sense of [26]): an output y is segmented by a fixed procedure D into units (sentences or atomic claims) $u_1, \dots, u_m$.

Definition 2.1 (Output measure)**.** $\hat{\mu}_y = \frac{1}{m}\sum_{j=1}^{m} \delta_{E(u_j)} \in \mathcal{P}(\mathbb{R}^d)$.

Assumption 1 (A1)*.* $\|E(u)\| \le R$ for all units (normalized embeddings: $R = 1$).

Assumption 2 (A2)*.* All kernels are bounded ($\sup_x k(x,x) \le K$), measurable, and characteristic on the relevant domain.

Assumption 3 (A3)*.* Calibration and test outputs are exchangeable as raw texts. Used only where stated; §4.4 removes it.

A *gate* is a pair (f_i, τ_i) with measurable $f_i : \Sigma^* \to [0, \infty)$ and threshold $\tau_i > 0$; write $\rho_i(y) = f_i(y)/\tau_i$, and gate i passes iff $\rho_i(y) \le 1$. The *instrument* $I = (D, \mathrm{NLI}, E, \text{graph builder})$ is fixed before calibration and treated as part of each f_i; this matters for Prop. 4.1. No manifold, bundle, or Fisher-metric structure is assumed; everything downstream lives on $(\mathbb{R}^d, \|\cdot\|)$ and on finite graphs.

3 The Five Gates

3.1 Format gate f_1

Let the structural specification be a formal language $L(G)$ (context-free grammar, JSON-Schema compilation, or regular constraint set). Define

$$f_1(y) = \frac{d_{\mathrm{edit}}(y, L(G))}{|y|}, \qquad d_{\mathrm{edit}}(y, L) = \min_{z \in L} d_{\mathrm{Lev}}(y, z).$$

For CFGs this is computable exactly by error-correcting Earley/CYK parsing [1] in $O(|G|^2|y|^3)$; for regular L, by product-automaton shortest path in $O(|A||y|)$. The gate is deterministic with zero estimation error, and $f_1(y) = 0 \iff y \in L(G)$. No projection onto a set of "well-formed" distributions is needed: an exact combinatorial minimization with a known algorithm does the work.

3.2 Fact gate f_2

Decompose y into claims $c_1, \dots, c_m$; retrieve evidence E_j from a verified knowledge base $\mathcal{K}$; let s_j be the calibrated entailment probability from an NLI model (isotonic, Platt, or temperature calibration on held-out data [16]; ECE reported). Per P3,

$$f_2(y) = 1 - \min_{1 \le j \le m} s_j,$$

with a mandatory *abstention state*: when retrieval confidence for a claim falls below a floor, the output is routed to abstention rather than scored—an unverifiable claim must not receive a numerical pass. (Kernel smoothing of a reference measure would instead interpolate silently over missing knowledge, returning a passing score where the knowledge base is simply empty.) Optionally, against a reference sample from $\mathcal{K}$ one may use the Sinkhorn divergence $S_\varepsilon(\hat{\mu}_y, \hat{\nu}_{\mathrm{fact}})$ [9], which metrizes weak convergence with dimension-robust sample complexity $O(n^{-1/2})$ [11]; plug-in W_2 degrades as $O(n^{-1/d})$ [10] and is therefore not used anywhere in this paper.

3.3 Logic gate f_3: weighted sheaf Laplacian, unified energy, relax-and-certify

Claim graph. $G = (V, E)$: vertices are claims; an edge (u, v) with relation type (entail / equiv / contradict) is added when the NLI model flags it, carrying its calibrated confidence $w_e \in [0, 1]$.

Weighted sheaf. Attach to G a cellular sheaf $\mathcal{F}$ [17]: stalks $\mathcal{F}(v) = \mathbb{R}^k$ (truth coordinates of claim v); restriction maps encode the relation (identity for entailment and equivalence, the negation lift $t \mapsto 1 - t$ for contradiction). With coboundary $\delta : C^0 \to C^1$ and $W = \mathrm{diag}(w_e)$, the weighted sheaf Laplacian is $L_{\mathcal{F}}(w) = \delta^\top W \delta$, and the raw score of an assignment x is $\langle x, L_{\mathcal{F}}(w) x\rangle^{1/2}$: a finite-dimensional Hodge-type quantity, computable in $O(|V|^3)$ worst case and by sparse solvers in practice. Global sections $\ker \delta = H^0(G; \mathcal{F})$ are exactly the edge-consistent assignments, so the score measures the obstruction to gluing the asserted claims into a global section—a cohomological reading carried here by a specified finite object rather than by analogy.

Proposition 3.1 (Lipschitz dependence on NLI confidences)**.** *If $\|x_v\| \le R$ and restriction maps have operator norm at most 1, then for confidence vectors w, w',*

$$\left|\langle x, L_{\mathcal{F}}(w) x\rangle - \langle x, L_{\mathcal{F}}(w') x\rangle\right| \le 4R^2 \|w - w'\|_1.$$

Proof. The energy is $\sum_e w_e \|(\delta x)_e\|^2$ with $\|(\delta x)_e\| \le 2R$; subtract termwise. $\square$

An out-of-distribution NLI perturbation therefore produces a bounded, quantified score perturbation, which composes with margin stability (Prop. 5.7); "garbage in, garbage out" is replaced by a sensitivity bound. The bound is verified numerically in the reference implementation.

Unified logic energy. Propositional constraints enter as sheaf data: lift Boolean assignments to $[0,1]$ stalks on the factor graph of extracted constraints, with restriction maps encoding literal polarity; let X be the linear-relaxation polytope of the clauses. Define the (pairwise) relaxed energy

$$E_{\mathrm{pair}}(y) = \min_{\substack{x\in X \\ x\equiv 1 \text{ on asserted claims}}} \langle x, L_{\mathcal{F}}(w)\, x\rangle^{1/2}.$$

The objective is a convex piecewise quadratic; projected gradient or interior-point methods find the global minimum. When a numeric extraction channel is available (equation chains with value flow), the same construction on affine restriction data yields a cycle-holonomy energy $E_{\mathrm{num}}(y)$: on a spanning tree the asserted value deltas propagate potentials, and each non-tree edge contributes its weighted closure gap (length-invariant and localized at the closing edge).

Definition 3.2 (Channel aggregation; max-form f_3)**.** *With channel thresholds $\tau_{\mathrm{pair}}, \tau_{\mathrm{num}}$ calibrated on clean data (split conformal per channel),*

$$f_3(y) = \max\Big(\tfrac{E_{\mathrm{pair}}(y)}{\tau_{\mathrm{pair}}},\ \tfrac{E_{\mathrm{num}}(y)}{\tau_{\mathrm{num}}}\Big).$$

The aggregation form is forced by the same dilution phenomenon as Prop. 3.7 one level down: additive (quadrature) combination lets pairwise-NLI noise on ordinary text drown the compositional channel. In the field study of §6.2, the quadrature form scores AUROC 0.586 *on compositional violations and the max-form* 0.990 from the same channel values*. Per-channel validity composes by the union bound (Theorem 4.3 applied channel-wise).*

Proposition 3.3 (Relaxation gap)**.** *(i) If the asserted constraint system is Boolean-satisfiable then* $E_{\mathrm{pair}}(y) = 0$*. (ii) The converse fails in general: LP/SDP integrality-gap instances (e.g. MAX-2-SAT [14]) yield fractional zeros for unsatisfiable systems.*

Proof sketch. (i) A Boolean satisfying assignment lies in X and has zero residual on every edge by construction of the restriction maps. (ii) Standard gap instances embed directly. □

Proposition 3.4 (No exact polynomial unification)**.** *A polynomial-time computable continuous score s with $s(y) = 0$ iff the asserted propositional system is Boolean-satisfiable would decide SAT in polynomial time. Hence, unless* P = NP*, every polynomial unified score is incomplete on the hard fragment.*

Proposition 3.5 (Structural equivalence on assertion-only acyclic graphs)**.** *Let G be a claim graph in which every vertex is asserted (clamped to 1) and whose edges are entailments and contradictions with weights w_e; let $C(G)$ denote its contradiction edge set. Then*

$$E_{\mathrm{pair}}(y)^2 = \sum_{e\in C(G)} w_e,$$

i.e. the relaxed unified energy is a strictly monotone function of the weighted contradiction sum (WSUM) and induces identical rankings (identical AUROC and identical false-positive rate at every operating point).

Proof. With every vertex clamped there is no free variable. Entailment residuals $\max(0, x_u - x_v)$ vanish at $x \equiv 1$, and each contradiction residual is $x_u + x_v - 1 = 1$; the constrained minimum is attained at the unique feasible point. □

Remark 3.6 (Where the sheaf can add capability). Prop. 3.5 delimits the tool's value: any detection separation between the sheaf energy and weighted counting requires *structure*—free (non-asserted) vertices whose values the global minimization can reconcile, cycles in the numeric/affine view whose holonomy no single edge witnesses, or the certificate layer. The field study of §6.2 confirms both directions of this boundary empirically.

Relax-and-certify. Consequently a SAT/SMT certifier is retained—not as a parallel gate but as a *certification layer*: it verifies (rounds) near-zero relaxed optima and, on UNSAT, returns a deletion-minimal unsatisfiable core whose size is an exact, interpretable severity overriding a passing continuous score. The soft/hard dichotomy of earlier drafts dissolves into the standard relax-and-certify hierarchy of modern optimization. As a diagnostic, the edges carrying the largest weighted residual energy localize the inconsistency; localization is evaluated in §6.2. A lattice-valued unification via the Tarski Laplacian [12] is recorded as Open Problem 8.1.

3.4 Alignment gate f_4

$f_4(y) = \sigma^{-1}(\hat{p}_{\text{unsafe}}(y))$: the logit of a safety classifier calibrated on held-out labels (ECE reported), composed with the conformal layer of §4 so that a bounded false-rejection rate on safe outputs holds as a theorem. *Computational independence* (structural requirement): f_4 operates on the raw text y and does not factor through D, the claim graph, or format parsing; a format-corrupted output therefore cannot even computationally block its own safety evaluation. Combined with the veto of §5.2, the masking channel is closed at the dataflow level as well as the decision level. (A W_2 projection onto a "compact convex set of safe behaviors" would be the natural distributional analogue, but no construction of such a set is known and metric projection in positively curved Wasserstein space is not well-posed; the calibrated classifier avoids both difficulties.)

3.5 Intent gate f_5

Proposition 3.7 (MMD dilution). *Let $\hat{\mu}, \hat{\mu}'$ be empirical measures on m units differing in exactly one unit, with kernel bound $k \leq K$. Then* $\mathrm{MMD}_k(\hat{\mu}, \hat{\mu}') \leq 2\sqrt{K}/m$.

Proof. $\psi(\hat{\mu}) - \psi(\hat{\mu}') = \frac{1}{m}(\phi(e_j) - \phi(e'_j))$ and $\|\phi(\cdot)\| \leq \sqrt{K}$. □

A fatal single-unit violation (one inserted negation) moves an aggregate-MMD score by $O(1/m)$—vanishing in long outputs. Hence, per P3,

$$f_5(y) = \max\bigl(f_5^{\text{hard}}(y),\, f_5^{\text{soft}}(y)\bigr).$$

Hard component (exact). Compile the instruction into programmatically checkable constraints $\{\kappa_\ell\}$ (length, inclusion/exclusion, ordering—the IFEval regime [31]); $f_5^{\text{hard}} = \max_\ell \mathbf{1}[\kappa_\ell \text{ violated}]$. A single negation that flips a checkable constraint scores 1 regardless of m. *Soft component (worst-unit, directed Hausdorff in feature space).*

$$f_5^{\text{soft}}(y) = \max_j \min_{r \in \mathrm{supp}(\nu_{\text{intent}})} \bigl\|\phi(E(u_j)) - \phi(E(r))\bigr\|_{\mathcal{H}_k},$$

where ν_{intent} is the reference measure of the task objective.

Proposition 3.8 (Unit-level sensitivity)**.** *If one unit lies at feature distance at least Δ from the entire intent reference set, then $f_5^{\text{soft}} \geq \Delta$, independently of m.*

The kernel mean embedding $\psi(\mu) = \int \phi \, d\mu$ is isometric with respect to MMD by construction, and MMD is a metric for characteristic k [15]. The isometry asserted here is with respect to MMD, not W_2: an isometric embedding of the W_2 geometry itself into a Hilbert space is impossible for $d \geq 3$ [2], so no such claim is made. Three honesty notes: (i) cross-encoder relevance scores are generally not positive semidefinite and may be used as raw scores, not kernels—conformal calibration is agnostic to the score's provenance; (ii) the $O(n^{-1/2})$ MMD concentration bound applies to the mean-embedding form; the max-form relies on conformal calibration for validity and on §7 for empirical power; (iii) aggregate MMD is retained solely as a drift-monitoring feature (§4.4), where distribution-level sensitivity is exactly what is wanted.

4 Calibration: Finite-Sample Guarantees Under Three Regimes

4.1 The guarantee is instrument-inclusive

Proposition 4.1 (Composite conformal validity)**.** *Fix the instrument I before calibration and let $f_i = g_i \circ I$. If calibration and test outputs are exchangeable as raw texts (A3), then Theorem 4.2 below holds verbatim for the composite score.*

Proof. Exchangeability is preserved under a fixed measurable map applied identically to every element of the sequence; the rank argument of Theorem 4.2 then applies to the composite scores unchanged. □

Claim-extraction truncation and out-of-distribution NLI behavior therefore cannot inflate the false-rejection rate: instrument noise enters calibration and test exchangeably. What instrument error does affect is detection power (Type-II), quantified in §4.5 and audited in §7.

4.2 Regime I — exchangeable deployment: split conformal

With a calibration set of n_i outputs certified valid on dimension i, set τ_i to the $\lceil (n_i+1)(1-\alpha_i) \rceil$-th smallest calibration score; see [3] for the construction in general form.

Theorem 4.2 (Distribution-free false-rejection control)**.** *Under (A3), for a new valid output y, $\mathbb{P}(\rho_i(y) > 1) \leq \alpha_i$; if scores are almost surely distinct, also $\mathbb{P}(\rho_i(y) > 1) \geq \alpha_i - \frac{1}{n_i+1}$.*

Proof. By exchangeability, the rank of $f_i(y)$ among the $n_i + 1$ scores is uniform on $\{1, \ldots, n_i + 1\}$ (ties only help the upper bound). Exceedance of τ_i requires the rank to exceed $\lceil (n_i+1)(1-\alpha_i) \rceil$, an event of probability at most α_i; the lower bound follows from the same rank computation without ties. □

Distribution-free false-rejection control thus arrives as a three-line finite-sample theorem, with no distributional assumptions. (Extreme-value refinements remain optional; note that all gate scores here are bounded, placing them in the Weibull, not Fréchet, max-domain of attraction [8]—any EVT use must carry domain diagnostics.) The field-study coverage audit (§6.3) verifies the exceedance bound on real chains ($0.095 \leq \alpha = 0.10$; three independent runs). The coverage-*distribution* refinement $\text{Beta}(m, n+1-m)$ describes the *exact conditional* coverage, which no finite audit observes directly: empirical per-split coverages convolve it with test-set binomial noise, split dependence under calibration-pool reuse, and tie discretization, so a naive KS test against the Beta law is miscalibrated—it rejects even ideal continuous i.i.d. scores, while the analytically exact conditional coverage passes (KS $p = 0.64$; null checks released with the code). Audits of the refinement must test against the binomially convolved law; the tie-safe exceedance bound is the deployable field statement.

Theorem 4.3 (Family-wise validity, safety-first)**.** *Choosing* $\alpha_i = \alpha/5$ *yields* $\mathbb{P}(\exists i : \rho_i(y) > 1 \mid y \text{ fully valid}) \leq \alpha$ *by the union bound. Where a fixed-sequence procedure is used to recover power under dependence, gates are tested safety-first (f_4 at full level α before all others), aligning power allocation with the veto semantics of §5.2.*

4.3 Regime II — estimable covariate shift: weighted conformal

When the shift is estimable (e.g. a changed prompt mix), weighted conformal prediction [27] restores the per-instance guarantee with likelihood-ratio weights $\hat{w}(x)$; weight-estimation error enters as an explicit total-variation correction. (Online density-ratio estimation as in [7] is a compatible single-gate instance.)

4.4 Regime III — arbitrary/adversarial drift: adaptive conformal inference

Deployment default: the online update of Gibbs and Candès [13], $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$, with the conformal quantile taken at level $1 - \alpha_t$.

Theorem 4.4 ([13], restated)**.** *For arbitrary—including adversarially chosen—sequences of distributions,*

$$\left| \tfrac{1}{T} \textstyle\sum_{t \leq T} \mathrm{err}_t - \alpha \right| \leq \tfrac{\max(\alpha_1, 1-\alpha_1)+\gamma}{\gamma T} \to 0.$$

Long-run false-rejection frequency converges to nominal α with no exchangeability or stationarity assumption; the guarantee does not lapse between recalibrations because adaptation is per-step—there is no "blind window" at the level of long-run frequency. (A3) is thereby demoted from a load-bearing assumption to the condition under which the stronger per-instance bound of Theorem 4.2 additionally holds.

Monitoring and fail-safe posture. Drift monitoring uses conformal test martingales / e-processes [28]: anytime-valid, no scheduled testing lag, false-alarm control at every stopping time; the monitored features include the aggregate-MMD statistic of §3. Upon alarm and until re-validation, thresholds are tightened monotonically and low-margin outputs are routed to abstention or human review.

Proposition 4.5 (Safe-side degradation)**.** *Decreasing τ_i enlarges every rejection region and shrinks none; during the alarm state the probability of a false accept on any dimension is non-increasing relative to the pre-alarm configuration.*

Proof. $\{f_i > \tau_i\}$ is decreasing in τ_i under set inclusion. □

What is sacrificed under drift is throughput (more abstentions), never the safety direction of the error.

4.5 Power accounting

Proposition 4.6 (Detection power lower bound)**.** *If an inserted violation survives claim decomposition with probability at least* $1 - \varepsilon_D$*, is flagged as an edge by NLI with probability at least* $1 - \varepsilon_{\text{NLI}}$ *given survival, and, once in the graph, drives* $\rho_3 > 1$ *with probability at least* π_3*, then*

$$\mathbb{P}(\textit{detected}) \geq (1 - \varepsilon_D)(1 - \varepsilon_{\text{NLI}})\,\pi_3.$$

ε_D and ε_{NLI} are estimated on the audit benchmarks of §7 with Clopper–Pearson intervals and enter the reported power certificate. *Abstention rule:* if extraction coverage (fraction of tokens attributable to extracted claims) falls below a floor $c_{\min}$, the output is routed to abstention.

4.6 Reporting requirements

For every deployment: n_i; empirical coverage on held-out valid data with Clopper–Pearson intervals; ECE for every internal classifier (f_2's NLI, f_4's safety classifier); the e-process monitor state; and the regime (I/II/III) under which each guarantee is claimed.

5 Decision Architecture: Partition, Veto, Stability

5.1 Parallel evaluation and the partition theorem

All five scores are computed for every output, unconditionally; no gate's computation may be short-circuited by another's failure. The primary logged object is the *failure vector* $K(y) = \{i : \rho_i(y) > 1\}$.

Lemma 5.1 (Measurability)**.** *Each* f_i *of §3 is measurable:* f_1 *is computable;* f_2, f_4 *are compositions of measurable maps with continuous links;* f_3 *is continuous in the assignment for each fixed graph, with graph extraction a measurable map into a countable set of structures;* f_5 *is a max/min of continuous functionals. Hence all maps below are measurable.*

Theorem 5.2 (Completeness and mutual exclusivity)**.** *For any finite family of measurable gates, with* $\Pi(y) = \min K(y)$ *(and* 0 *if* $K(y) = \varnothing$*),*

$$\Sigma^* = V_{\text{valid}} \sqcup \bigsqcup_{i=1}^{5} E_i, \qquad E_i = \{y : \rho_i(y) > 1,\ \rho_j(y) \leq 1\ \forall j < i\}.$$

Proof. Well-ordering of the failure set $K(y)$. □

Remark 5.3 (Honesty). Theorem 5.2 holds for any five measurable criteria and certifies nothing about gate quality; it is bookkeeping. The substantive guarantees are Theorems 4.2–4.4 and the propositions of this section.

5.2 Safety veto and the demotion of Π

$$\mathrm{Dec}(y) = \begin{cases} \textbf{Block} & 4 \in K(y) \quad \text{(safety veto, order-independent)} \\ \textbf{Repair via } \Pi_{-4}(y) & K(y) \neq \varnothing,\ 4 \notin K(y) \\ \textbf{Pass} & K(y) = \varnothing \end{cases}$$

where $\Pi_{-4}(y) = \min K(y)$ over the non-veto gates. Π survives only as a repair-routing heuristic ("fix the format before re-checking facts"), a role in which lexicographic order is defensible; it carries no reporting or risk semantics.

Proposition 5.4 (No masking)**.** *Under* Dec*, every output with* $\rho_4(y) > 1$ *is Blocked regardless of* $\rho_1, \rho_2, \rho_3, \rho_5$*. The scenario "*$f_1$ *fails first and suppresses an* f_4 *violation" is unrealizable: "first" does not exist at the decision layer, and by §3 it does not exist at the dataflow layer either.*

Remark 5.5 (Π is not a security boundary). Consider an adversary who deliberately plants a mild fact violation so that f_2 "fails first." Against a sequential, short-circuiting evaluator this attack masks a co-occurring alignment violation—it is valid against any sequential, short-circuiting semantics, including the lexicographic-only design this architecture replaces. Under the present architecture it accomplishes nothing: all five scores are computed unconditionally, $K(y)$ records both failures, f_4 reads the raw text, and Dec blocks on $4 \in K(y)$ irrespective of every other entry. Mutual exclusivity is a property of the reporting taxonomy Π (one primary label per output); diagnostic completeness is carried by $K(y)$, which is always full. The security boundary is Dec, never Π.

5.3 Well-posedness and stability

Proposition 5.6 (Almost-sure well-posedness)**.** *If each* $f_i(y)$ *has an atomless law at* τ_i *under the deployment distribution, then almost surely every output has strictly positive margin* $\gamma(y) = \min_i |f_i(y) - \tau_i|$*, and* Π*,* K*,* Dec *are locally constant under score perturbations smaller than* $\gamma(y)$*.*

The condition is empirically checkable via continuous score histograms. We note what it deliberately avoids: stating well-posedness as a boundary-regularity condition on the space of measures would require a canonical reference measure on Wasserstein space, of which there is none, and a Hausdorff-dimension bound, which is meaningless in infinite dimensions.

High dimensions and measure concentration. The classification factors through the five scalar scores: Dec depends on y only via $(f_1(y), \ldots, f_5(y)) \in \mathbb{R}^5$. High-dimensional geometry of the embedding space therefore enters only through the one-dimensional law of each score; no claim about the Hausdorff dimension or measure of a boundary in $\mathcal{P}(\mathbb{R}^d)$ is made or needed. The legitimate residue of the concentration concern is empirical: score laws might concentrate near τ_i, producing many small-margin outputs. Two answers. First, the validity guarantees (Theorems 4.2 and 4.4) hold regardless of concentration; margins matter only for estimation stability (Prop. 5.7). Second, we mandate reporting the empirical margin distribution $\hat{\mathbb{P}}(|f_i - \tau_i| \le \delta)$ and support an optional *abstention band*: outputs with margin below δ are routed to review, converting boundary ambiguity from a silent failure mode into a measured abstention rate.

Proposition 5.7 (Margin-based stability of estimated classification)**.** *If simultaneously* $|\hat{f}_i - f_i| \le \varepsilon_i$ *for all i with probability at least* $1 - \delta$ *(MMD concentration for the mean form;* $\varepsilon_1 = 0$*; Prop. 3.1 converting NLI-confidence error into* ε_3*) and* $\gamma(y) > \max_i \varepsilon_i$*, then* $\widehat{\mathrm{Dec}}(y) = \mathrm{Dec}(y)$ *with probability at least* $1 - \delta$*.*

Remark 5.8 (Stability–sensitivity separation). Two perturbation classes must not be conflated. *Measurement* perturbations (sampling noise, NLI-confidence noise) should not flip decisions; Prop. 5.7 delivers this above the margin. *Semantic* perturbations (inserting a single negation) should flip decisions, and the gates are engineered to be non-smooth in exactly this direction: a single-unit flip at feature distance Δ moves f_5^{soft} by at least Δ (Prop. 3.8) and moves the min/max aggregations of f_2 and f_5^{hard} by $O(1)$, while measurement noise moves scores by at most ε; the two regimes are distinguishable whenever $\Delta > 2\varepsilon$. Local constancy is claimed in the score metric, never in the semantic metric; demanding the latter would reintroduce the dilution pathology that Prop. 3.7 rules out.

Proposition 5.9 (One-sided metric comparison)**.** *For L-Lipschitz bounded k and measures supported in the R-ball,* $\mathrm{MMD}_k \le C(L,K)\, W_1 \le C(L,K)\, W_2$*. The converse inequality fails in general; only this direction is used, and only for interpretability (small transport perturbations of an output cannot inflate the soft intent score), never for a guarantee.*

A global Lipschitz equivalence of the W_2, harmonic, and RKHS norms would be convenient but is false as stated; only the one-sided direction above is available, and the paper uses no more than that.

5.4 Computational profile and anytime operation

Latency is graded per gate, and the architecture guarantees that safety never waits for expensive computation. *Fast path (synchronous, per output):* f_4 is one classifier forward pass on the raw text, and f_1 is linear-time validation (membership checking), with the cubic error-correcting parse invoked only on failure to quantify severity; the veto of §5.2 therefore executes within a single forward-pass latency. *Slow path (deadline-bounded):* f_2 is dominated by retrieval and NLI passes; the relaxed energy of f_3 is a sparse convex program on a claim graph with m vertices (m in the tens for typical outputs), i.e. millisecond-scale linear algebra, and is updatable incrementally under streaming generation by low-rank updates of $L_{\mathcal{F}}(w)$ as claims and edges arrive. The SAT certifier is invoked lazily under a deadline: the relaxed score is an anytime answer, and if the deadline expires the output carries an "uncertified" flag with conservative handling. This is a principled rather than an apologetic design: Prop. 3.4 shows the worst-case certification cost is unavoidable for any exact method, so the only honest options are anytime relaxation plus optional certification—which is what the gate does. *Scope honesty:* microsecond decision loops (e.g. high-frequency trading) are outside the scope of any semantic verification—no entailment model evaluates in microseconds; the target regime is the one in which generation itself costs 0.1–10 s and full verification runs at a fraction of that budget, with the safety fast path always inside it.

6 Mechanism Study and Pre-Registered Field Study: Sheaf Energy versus Contradiction Counting

The framework's mathematical identity rests on one empirical question: does the sheaf energy f_3 add signal beyond the trivial baseline that counts NLI-flagged contradiction edges above a decision threshold? This section answers it in two stages: a simulation that isolates the *mechanism* by which a separation could arise (§6.1), and the pre-registered *field study* that executed the protocol with a real, calibrated instrument on real reasoning chains (§6.2). The field study adjudicates the simulation's assumptions, confirms the pre-registered success criteria where the separation is structural, and sharpens the claim into Prop. 3.5.

6.1 Mechanism simulation

The simulation isolates the mechanism under two documented assumptions about NLI behavior:

(S1) *Distance decay.* The confidence assigned by a pairwise NLI model to the direct contradiction edge decays with the semantic/reasoning distance k between the violating statement and the claim it contradicts; in the simulation, $c(k) = \mathrm{clip}(0.95 - 0.13k + \mathcal{N}(0, 0.06))$, falling below the standard 0.5 decision threshold around $k \ge 3$.

k	AUROC Count	AUROC Sheaf	AUROC Sat	Loc. Count	Loc. Sheaf
1	0.881	1.000	1.000	100.0%	100.0%
2	0.875	1.000	1.000	100.0%	100.0%
3	0.831	1.000	0.933	86.7%	100.0%
4	0.537	0.993	0.554	10.8%	97.5%
5	0.504	0.970	0.500	0.0%	83.3%

Table 1: Mechanism simulation: detection AUROC and top-1 localization rate versus violation distance k ($n = 120$ per cell), *under simulated confidences drawn from (S1)/(S2).* Threshold-based methods collapse to chance as the direct-edge confidence crosses the threshold; the weighted sheaf energy degrades gracefully and retains localization.

(S2) *Noise in both classes.* Hedged or hypothetical claims (considered and rejected by the text, hence not asserted) are flagged by NLI as contradicting asserted claims with confidence $w \sim U(0.55, 0.85)$; and spurious sub-threshold asserted–asserted contradiction edges occur with $w \sim U(0.10, 0.45)$.

Setup. Synthetic reasoning chains of six asserted claims connected by entailment edges ($w \sim \text{Beta}(20, 2)$); violated cases append an asserted statement contradicting claim A_0 at step distance $k \in \{1, \dots, 5\}$, with the direct edge confidence drawn per (S1) and an optional weak paraphrase edge to A_1; both classes receive (S2) noise. 120 cases per (k, class) cell. Three detectors: Count (#contradiction edges with $w \geq 0.5$; the standard baseline), Sheaf (the unified energy of §3.3, assertion-aware, no thresholding), Sat (DPLL on edges with $w \geq 0.5$).

The simulation establishes a conditional statement: *if* real NLI confidences decay with distance (S1) and produce hedged-claim false positives (S2), *then* the sheaf energy separates structurally from counting (Table 1). Both assumptions are directly testable, and the harness becomes the real experiment by replacing simulated confidences with model outputs. That experiment follows.

6.2 Pre-registered field study

Design (protocol frozen before data collection; deviations logged). Instrument: DeBERTa-large [18] fine-tuned on MultiNLI [29], temperature-scaled [16] on MNLI validation-matched ($n =$ 2,000; $T = 1.452$; 15-bin ECE 0.011 post- vs. 0.040 pre-calibration; MNLI accuracy 0.911); the instrument never receives injection locations. Corpus: 450 GSM8K [6] gold solutions parsed into 4–10-step claim chains, with the dataset's native calculator annotations as the numeric extraction channel. Paired clean/violated design, $n = 150$ cases per cell (45 for per-channel threshold calibration, 105 for evaluation); 22,706 scored sentence pairs. Violation families: **V1/V2**, template negation of the step $k \in \{1, \dots, 5\}$ positions before the end, appended at the end; **V3**, a compositional numeric violation—a closing sentence misstates the ratio between final and initial computed quantities while every sentence *pair* remains individually consistent (pairwise detectors blind by construction); **V4**, hedged contamination attached as a non-asserted vertex in both classes. Arms: Count, Wsum (de-thresholded weighted sum—the control that prevents attributing a sheaf win to mere de-thresholding), NliMax, Sat, and the max-form sheaf energy of Def. 3.2. Primary endpoints: paired-bootstrap ΔAUROC against the best baseline per cell with Holm correction across cells; top-1 localization. Pre-registered success criteria: ΔAUROC ≥ 0.03 (CI excluding 0) or localization advantage ≥ 0.10, plus a Wsum-specific win on V3 or V4.

Cell	$\textsc{Sheaf}_{\text{max}}$	$\textsc{Sheaf}_{\text{quad}}$	Count	Wsum	NliMax	Sat	Loc. S/B
V1 k=1	0.719	0.719	0.696	0.719	0.531	0.514	0.21 / 0.94
V2 k=2	0.741	0.741	0.710	0.741	0.520	0.505	0.17 / 0.91
V2 k=3	0.756	0.756	0.734	0.756	0.513	0.505	0.06 / 0.94
V2 k=4	0.717	0.717	0.685	0.717	0.503	0.500	0.03 / 0.90
V2 k=5	0.739	0.739	0.701	0.739	0.502	0.500	0.02 / 0.92
V3	**0.990**	0.586	0.498	0.501	0.500	0.500	**0.24 / 0.00**
V4 k=1	0.751	0.751	0.696	0.732	0.518	0.510	0.14 / 0.91

Table 2: Field study (confirmatory run; $n = 105$ evaluation cases per cell). AUROC per arm; Loc. S/B is top-1 localization for the sheaf hotspot vs. the baseline above-threshold rule. On V3 the sheaf–best-baseline gap is $\Delta = +0.489$ [95% CI $0.470, 0.503$], Holm-adjusted $p = 0.0018$, with FPR at 95% TPR of 1.0% (vs. 94.3% for every pairwise baseline); the Wsum anti-artifact clause holds. V1/V2 equalities are exact (Prop. 3.5).

Assumption adjudication (E0). Both simulation assumptions *fail* in the field, in informative ways. (S1): calibrated contradiction confidence on the direct pair $(\neg c_i, c_i)$ is *flat* in k $(0.749, 0.740, 0.742, 0.764, 0.756$ for $k = 1, \ldots, 5)$—a modern NLI recognizes a verbatim negation regardless of its position; cross-pair confidence sits near 0.5 with a mild downward drift. (S2): hedged constructions were flagged in 0/300 trials (Clopper–Pearson 95%: $[0, 0.012]$). Consequently the collapse mechanism of Table 1 is not exercised by verbatim negations, and hedged reconciliation has nothing to discount under this instrument.

Findings (Table 2). Three results delimit the tool's value precisely.

(i) The structural regime: confirmed at 0.990*.* On compositional violations (V3), every pairwise arm is at chance by construction—no sentence pair witnesses the inconsistency—while the max-form sheaf energy detects the broken cycle holonomy at AUROC 0.990 with 1% FPR at 95% TPR, and is the only method with nonzero localization (0.24 vs. exactly 0). The pre-registered criteria are met with the Wsum clause, so the separation is not a de-thresholding artifact.

(ii) The structure-free regime: provable coincidence. On V1/V2, the sheaf and Wsum columns agree to three decimals in every cell—an instance of Prop. 3.5, not an empirical accident. Field NLI noise (spurious contradiction edges between ordinary solution steps) caps all pairwise methods at $\approx$0.72–0.76 and drives Sat to chance; sheaf top-1 localization loses to the simple above-threshold rule outside V3 for the same reason. The localization claim is therefore scoped to compositional violations.

(iii) The aggregation form is forced. The as-specified quadrature combination of the two f_3 channels scores 0.586 on V3; the max-form of Def. 3.2 scores 0.990 from the same channel values. The dilution phenomenon of Prop. 3.7 recurs inside f_3, and its P3 remedy applies one level down.

6.3 Field test of the calibration layer

On clean-chain sheaf scores (pool $n = 735$; $n_{\text{cal}} = 200$, $\alpha = 0.10$, $R = 1{,}000$ random splits), pooled exceedance is $0.0952 \leq \alpha$ (exact binomial $p = 1.0$ against inflation): the finite-sample bound of Theorem 4.2 holds in the field (and again in the replication and scale-up runs: 0.0995 and 0.0996). A naive KS test of per-split coverages against the $\text{Beta}(m, n+1-m)$ refinement rejects ($p \approx 0$), but null checks show this reflects the audit statistic, not the law: the same KS rejects ideal continuous i.i.d. scores under the same procedure (test-set binomial noise, pool reuse, and, here, 39.7% tied scores), while the analytically exact conditional coverage passes (KS $p = 0.64$). See §4 for the corrected audit prescription; the tie-safe exceedance bound is the field-verified object.

Scale-up. A doubled-scale run (1,000 chains, $n = 210$ evaluation cases per cell, six negation and three closing-sentence template families, 46,936 scored pairs) reproduces every finding: V3 at AUROC 0.995 ($\Delta = +0.488$ $[0.474, 0.498]$, FPR at 95% TPR 0.5%, localization 0.30 vs. 0.00), exact V1/V2 counting-coincidence, quadrature dilution (0.576), and the Regime-I bound ($0.0996 \leq 0.10$, pool $n =$ 1,470). The headline is therefore stable across sample size and template families.

Replication with a second instrument. The key cells were replicated with an architecture-distinct instrument (RoBERTa-large [23] fine-tuned on MNLI, $T = 1.337$, ECE 0.012; 600 chains; $n = 150$/cell). The headline replicates: on V3, the max-form sheaf energy scores AUROC **0.999** (FPR at 95% TPR 0.5%; localization 0.33 vs. 0.00) against chance-level baselines ($\Delta = +0.489$ $[0.479, 0.498]$, Holm $p = 0.0008$); the quadrature dilution (0.593) and the structural coincidence on V1 (max-form $\equiv$ WSUM = 0.752) also replicate; the Regime-I bound holds again ($0.0995 \leq 0.10$). Instructively, this instrument's hedge false-positive rate is *nonzero* (2.3%, Clopper–Pearson $[0.9\%, 4.7\%]$), and exactly as the mechanism predicts, the assertion-aware reconciliation capability activates: on V4 the sheaf beats WSUM by +0.035 ($\geq$ the pre-registered 0.03 bar) where under the zero-S2 instrument it was vacuous. The V4 capability is thus an *instrument-conditional* one, switched on precisely when S2 > 0—a falsifiable dependence the instruments jointly demonstrate as a dose–response pattern: S2 = 0.7% (DeBERTa, scale-up) yields +0.015 over WSUM, S2 = 2.3% (RoBERTa) yields +0.035.

Scope of the claim. Violations are template-inserted into real chains (the pre-registered method); an organically generated error corpus is registered as the immediate next step. All code, the frozen protocol with its deviation log, and per-run artifacts are released with the paper.

7 Pre-Registered Empirical Protocol

The protocol below was registered, and frozen, before any field data were collected. Status markers record what the M1 field study has executed; the remainder is the active experimental agenda.

(1) Datasets and gate mapping. *Format:* structured-output benchmarks plus synthetic grammar corruption at controlled edit distances (ground-truth f_1). *Fact:* TruthfulQA [22] and FEVER-style verification with a frozen retrieval index. *Logic:* GSM8K/MATH step-consistency with adversarial contradiction insertion at known graph locations and controlled distance k. *[Executed for GSM8K in §6.2; MATH pending an audited equation extractor.]* *Alignment:* human-adjudicated safety benchmarks. *Intent:* IFEval-style verifiable-constraint suites.

(2) Primary metrics. Per-gate AUROC and FPR at 95% TPR; empirical conformal coverage versus nominal α with Clopper–Pearson intervals (a direct field test of Theorem 4.2) *[executed: §6.3]*; ACI long-run error tracking under injected drift (Theorem 4.4); ECE of internal classifiers *[executed for the NLI instrument]*; end-to-end Dec accuracy against human triage.

(3) Ablations with pre-registered predictions. (a) MMD vs. Sinkhorn vs. plug-in W_2: plug-in W_2 degrades with embedding dimension per its $n^{-1/d}$ rate; the others do not. (b) *Sheaf vs. counting vs. SAT-only on real chains—the empirical linchpin. [Executed: §6.2. Success criteria met on the compositional cell ($\Delta AUROC$ +0.489, Holm $p = 0.0018$, WSUM clause satisfied); the distance-collapse pattern of Table 1 does not transfer to verbatim negations because (S1) fails in the field; V4's mechanism is vacuous under a calibrated instrument because (S2) fails. The claim is re-scoped accordingly (Prop. 3.5, Remark 3.6).]* (c) Lexicographic Π vs. full $K(y)$ for repair routing.

(d) Mean-MMD vs. max-form f_5 on a single-negation adversarial suite: the mean form's miss rate tends to 1 as m grows, by Prop. 3.7.

(4) Assumption tests. Direct measurement of (S1) and (S2). *[Executed: both fail in the field; see §6.2. This is the protocol working as intended—the simulation's premises were falsifiable and have been adjudicated.]*

(5) Dependence diagnostics. Empirical correlation matrix of $(f_1, \ldots, f_5)$; strong dependence motivates the safety-first fixed-sequence procedure of Theorem 4.3.

(6) Instrument audit. Human-annotated decomposition benchmark for ε_D (claim-level recall, Clopper–Pearson interval); edge-detection benchmark with adversarial paraphrase/negation for $\varepsilon_{\mathrm{NLI}}$; abstention floor $c_{\min}$ reported with the resulting abstention rate; all feeding the power certificate of Prop. 4.6.

(7) Drift monitoring. E-process monitor state and recalibration events logged; coverage audited within drift episodes, where the Regime-III guarantee is exactly the testable object.

(8) Registered next steps. Scale-up (n = 1,000/cell) with template-family robustness; replication with an architecture-distinct instrument (RoBERTa-large-MNLI); an organic-error corpus (model-generated solutions with execution-labeled compositional errors) replacing template insertion for the headline claim.

8 Limitations and Open Problems

8.1 Instrument dependence. All distributional gates are relative to the fixed E, D; guarantees are for the scores as defined (Type-I end-to-end by Prop. 4.1; Type-II bounded by Prop. 4.6 and audited). Encoder-class invariance is open. The field study adds a concrete instance: V4's hedged-reconciliation capability is contingent on the instrument's hedge false-positive rate: zero for a calibrated DeBERTa-large-MNLI (capability vacuous) and 2.3% for a RoBERTa-large-MNLI (capability active, +0.035 over WSUM)—an instrument-conditional dependence the replication demonstrates rather than merely hypothesizes.

8.2 Assumption ledger. Regime I requires exchangeability; Regime II requires estimable shift; Regime III guarantees long-run frequency only. Per-instance guarantees under fully adversarial shift are impossible in general; the fail-safe posture (Prop. 4.5) is the stated conduct in that gap. Marginal (not conditional) coverage is delivered; exact conditional coverage is in fact unattainable distribution-free [4], so the designated next theoretical step is the achievable relaxation: coverage conditional on prompt-difficulty strata via Mondrian conformal. The field E2 audit adds: the coverage-distribution refinement is not directly auditable by a naive KS of empirical coverages (test-binomial noise, pool reuse, and score ties each break the comparison—null-checked in §6.3); the exceedance bound is tie-safe and held in all three runs.

8.3 Relaxation gap and certificate-layer fragility. A continuous zero of f_3 certifies only relaxed consistency (Prop. 3.3); the certifier closes the gap at worst-case exponential cost, and Prop. 3.4 shows this is unavoidable unless P = NP. The field study adds a caution on the certificate layer's *input*: hard clauses assembled by thresholding NLI edges at $w \geq 0.5$ inherit the instrument's false-positive edges, driving the thresholded SAT arm to chance on real chains (§6.2); certificates therefore certify the thresholded graph, not the text, and a calibrated edge-inclusion threshold (or weighted-clause certificates) is the registered repair before the certificate layer's field role can carry weight.

8.4 Scope of the field evidence. §6.2 uses template-inserted violations on a single corpus (GSM8K), now under two architecture-distinct instruments; the localization claim is scoped to compositional violations, and the V1/V2 regime is provably counting-equivalent (Prop. 3.5). An organic-error corpus and MATH remain registered next steps (§7(8)).

8.5 Lexicographic repair order. A convention; no optimality claim is made.

Open Problem (Tarski unification). Recast the hard and soft logic layers as sheaves valued in complete lattices with the Tarski Laplacian [12], where global sections are fixed points and Boolean semantics is native.

Open Problem (holonomy power theorem). In a random-chain model with a planted closing-claim violation of gap γ and extraction noise σ, derive the explicit power curve of the cycle-holonomy statistic as a function of γ/σ and the cycle's minimum edge confidence, upgrading the observed field separation (Table 2) into a predictive theorem.

9 Conclusion

Five well-posed, individually guaranteed instruments beat one undefined manifold. The framework presented here gives the guardrail architecture already running in production its missing mathematical contract: computable severities, finite-sample false-rejection control graded across three drift regimes (now field-verified in Regime I), a complete and mutually exclusive error taxonomy, an unmaskable safety veto, and—in the sheaf logic gate—a detection capability whose boundary is now a theorem plus a measurement: it provably coincides with weighted counting on structure-free claim graphs, and in its structural regime, adjudicated by a pre-registered field study, it detects compositional violations invisible to every pairwise method at AUROC 0.990 while uniquely localizing them.

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